

KHADEEJA HANNA P

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PROFESSIONAL SUMMARY

AI & Data Science graduate with hands-on experience building end-to-end AI applications, including RAG pipelines with vector databases, OCR-based document processing, and voice-driven conversational systems using Python, FastAPI, and Flask. Skilled in Machine Learning, Deep Learning, Computer Vision, and NLP, seeking an entry-level AI Engineer role.

EDUCATION

Bachelor of Technology in Artificial Intelligence and Data Science

MES College Of Engineering Kuttipuram - APJ Abdul Kalam Technological University

Oct 2022 – May 2026

Malappuram, Kerala

EXPERIENCE

AI/ML Intern

EdiGlobe

July 2026 – Present

Remote

- Performed data preprocessing and exploratory data analysis (EDA) using Python, Pandas, and NumPy
- Cleaned and prepared datasets by handling missing values, duplicates, and feature inconsistencies.
- Built classification models using Scikit-learn to support predictive analytics on structured datasets.

AI/ML Intern

Outrix

July 2025 – Aug 2025

Remote

- Analyzed datasets using Pandas and NumPy to identify patterns, trends, and actionable insights.
- Applied Python-based machine learning techniques to support real-world data analysis tasks across structured datasets.
- Developed and evaluated machine learning models using Scikit-learn, applying classification techniques to derive predictive insights from structured datasets.

PROJECTS

Mentora AI - AI-Powered Mentor and Tutor System

Python, Flask, SQLite, OCR, Twilio, Gemini API

- Built an AI mentoring platform integrating OCR, an LLM-based chatbot, and automated parent communication.
- Implemented an OCR pipeline using Groq LLaMA 4, achieving 90% data extraction accuracy from uploaded academic documents.
- Enabled multilingual (Malayalam/English) outbound parent calls via Twilio and Gemini, achieving a 90% call connection rate.

RAG Document Assistant - AI-Powered Document Q&A System

Python, React, FastAPI, ChromaDB, Sentence Transformers, Gemini API

- Built a Retrieval-Augmented Generation (RAG) pipeline supporting document uploads up to 5MB.
- Implemented ChromaDB vector storage and Sentence Transformers for semantic chunk embedding and retrieval.
- Developed a relevance-scoring mechanism that displays each retrieved chunk's relevance to the query as a percentage, improving result transparency.

AI-Based X-Ray Image Enhancement Using Real-ESRGAN

Python, PyTorch, Real-ESRGAN, OpenCV, Flask

- Collaborated in a 3-member academic team to develop a Real-ESRGAN super-resolution system using PyTorch for enhancing low-resolution X-ray images.
- Improved image quality by increasing PSNR from ~20 dB to 35+ dB and SSIM from ~0.45 to 0.90+.
- Built a Flask-based web interface for uploading low-resolution X-rays and displaying enhanced images with PSNR/SSIM quality metrics.

TECHNICAL SKILLS

AI/ML: RAG, Machine Learning, Deep Learning, NLP, Computer Vision, Gemini API, Sentence Transformers, Vector Databases

Programming & Libraries: Python, SQL, C/C++, Pandas, NumPy, Scikit-learn, PyTorch, OpenCV

Frameworks & APIs: FastAPI, Flask, REST APIs, Twilio API

Databases & Tools: MongoDB, PostgreSQL, MySQL, SQLite, ChromaDB, Git, GitHub, Docker, Jupyter Notebook, Power BI, Excel

ACHIEVEMENTS

- Awarded "Galactic Problem Solver" certificate at NASA International Space Apps Challenge 2024.

COURSES & CERTIFICATIONS

- Partnering with AI in the Workplace Job Simulation – Datacom, Forage, 2026
- Accessing Hadoop Data Using Hive – IBM Cognitive Class, 2025
- 1 Million Prompters - Dubai Centre for Artificial Intelligence & Dubai Future Foundation, 2026